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ORIGINAL PAPER

Protective effect of *Pueraria tuberosa* DC. embedded biscuit on cisplatin-induced nephrotoxicity in mice

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Abstract Recently, the nephroprotective property of *Pueraria tuberosa* DC. tuber (PT) has been reported by our group. Here, PT-embedded biscuits were prepared and tested on cisplatin-induced nephrotoxicity in Swiss albino mice. The PT powder was characterized by RAPD (random amplified polymorphic DNA) to ascertain its authenticity and PT biscuits were prepared in different concentrations (1, 2, or 4 g of PT powder). These biscuits were given as diet for a total of 10 days, but on the 7th day cisplatin injection (8 mg/kg bw, i.p.) was given. On the 10th day animals were killed to collect kidneys for assessment of antioxidant status. Blood samples were collected on both the 7th and 10th days for assessment of liver and kidney functions. In mice, PT biscuit showed significant protection against cisplatin-induced nephrotoxicity, but there was a transient rise in alanine aminotransferase and aspartate aminotransferase at the dose of 4 g PT biscuit. Therefore, it is suggested that PT biscuit might be an effective food supplement for cancer patients undergoing cisplatin-chemotherapy. However, periodical liver function monitoring is required, especially when PT is used for longer periods or at higher doses.

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Introduction

The prevalence of cancer is constantly increasing and radiotherapy and chemotherapy are established therapeutic interventions; both, however, are associated with several side effects. Drugs like cisplatin are known to have nephrotoxicity [1], and thus there is a need to develop herbal food supplements to manage this adverse effect of anticancer drugs.

Cisplatin-induced nephrotoxicity is a complex process and multiple mechanisms have been suggested, including caspase activation, mitochondrial dysfunction, DNA damage, inhibition of protein synthesis, activation of p53 tumor suppressor proteins, activation of the mitogen activated protein kinase (MAPK)-signalling pathway, oxidative stress and inflammation. All these mechanisms ultimately lead to tubular cell death [2–8].

Herbal remedies have become popular in recent years, because they are natural and eco-friendly [9]. *Pueraria tuberosa* DC. powder (PT powder) is in clinical use in Ayurvedic medicine in various formulations as a tonic, aphrodisiac, demulcent, lactogogue, purgative and cholagogue [10]. Some of its important biological properties include anti-hyperlipidemic [11], anti-fertility in male rats [12], anti-hyperglycemic [13] and hepato-protective [14]. Our earlier reports have indicated antioxidant [15] and anti-inflammatory properties [16], nephroprotectivity against cisplatin [8] and hepatotoxicity in higher doses [17] in methanolic extract of PT powder. Thus, because of the nephroprotective property of the methanolic extract, we decided to develop a medicinal biscuit containing PT powder.

Since PT tubers are collected from the wild and sold in markets, so it is mandatory to identify the actual plant before considering any biological experiment. Here, therefore, we used a RAPD (random amplified polymorphic DNA) method to identify the PT sample which was used in our experiments.

Materials and methods

Materials

Pueraria tuberosa powder, cisplatin injection (cytoplatin-10, Cipla), wheat flour, sugar, vanaspati ghee (fat), refined oil, vidarikhand powder, skimmed milk powder, vanilla powder, salt, water. Sodium bicarbonate, ammonium bicarbonate, sodium meta-bisulphite, emulsifiers, electric mixer with variable speeds, metal gauge strips (7 mm), rolling pin, biscuit cutter, baking oven, CTAB [cetyl trimethyl ammonium bromide, 20% (w/v)], 1 M Tris–HCl (pH 8), 5 M EDTA (pH 8), 5 M NaCl, 3 M sodium acetate, ethanol, chloroform–isoamyl alcohol [24:1 (v/v)], polyvinylpyrrolidone (PVP) (40,000 MW) (Sigma), and β-mercaptoethanol. All the chemicals used in the experiments were of analytical grade. Enzymes (*Taq* polymerase and RNase A), Taq buffer, $MgCl_2$ and dNTPs for polymerase chain reaction (PCR) amplification were purchased from Bangalore Genei (Bangalore, India).

Characterization of *Pueraria tuberosa* through RAPD

The tubers of *Pueraria tuberosa* were purchased from the local market and its authenticity was confirmed by comparison with the fresh sample collected from the ayurvedic garden, Department of Dravya-Guna, Institute of Medical Sciences, Bananas Hindu University, and also by direct comparison with the sample preserved in our department (voucher no. YBT/MC/12/1-2007). The dried tubers were coarsely powdered. For identification through RAPD, its DNA was extracted according to the protocol of Khanuja et al. [18] with slight modification. Briefly, plant material was ground in liquid nitrogen and added to freshly prepared extraction buffer [100 mM Tris–Cl (pH 8.0), 25 mM EDTA, 1.5 M NaCl, 3% CTAB, with 1% β-mercaptoethanol (v/v) and 1% PVP (w/v) (added immediately before use)], and incubated at 60°C in a shaking water bath for 2 h. Thereafter, this mixture was extracted with chloroform:isoamyl alcohol (24:1) at room temperature and the upper clear aqueous layer was saved to another tube. This was mixed with an appropriate volume of 5 M NaCl, to obtain a final concentration of 1.5 M, and isopropanol (0.6 volumes). It was kept at −20°C for 1 h and DNA fibers were spooled out, washed with 80% ethanol and dried overnight in vacuum desiccators. Next day it was dissolved in high-salt TE buffer [1 M NaCl, 10 mM Tris–Cl (pH 8.0) and 1 mM EDTA] and DNA concentrations were measured by reading the absorbance at 260/280 ratio. An aliquot of DNA preparation was mixed with loading buffer containing ethidium bromide and run on 1.2% agarose gel to assess its purity.

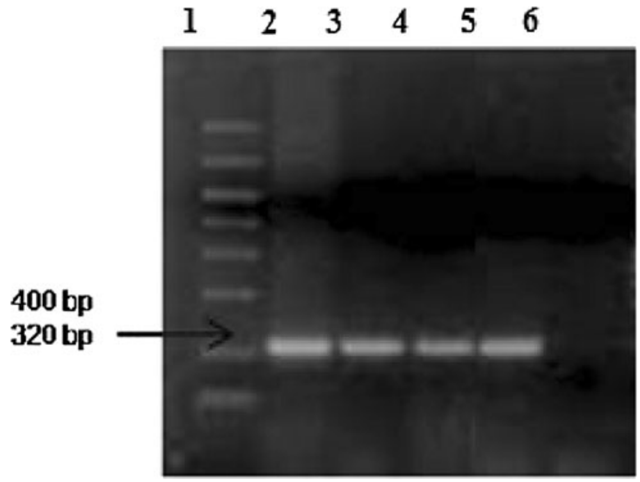

Fig. 1 PCR amplification of *P. tuberosa* using SCAR marker (PT 1 FP and PT 1 RP). Lanes: *1* molecular weight ladder, *2–5* amplified DNA of *P. tuberosa* tubers, obtained from various sources, *6* negative control amplification with genomic DNA of *I. mauritiana*

The RAPD reaction was performed according to the protocol of Devaiah and Venkatasubramanian [19]. A pair of SCAR (sequence characterized amplified region) oligonucleotide primers (Pt 1 FP; Pt 1 RP) was designed which could amplify ~320 bp of the genomic *P. tuberosa* DNA. Pt 1 FP (22-mer, forward primer AGAGCCAGAG TCCGCCAGGCCT) and Pt 1 RP (26-mer, reverse primer ACCAGCGAGCAATGAACCTCATCTTC), designed earlier [19], were commercially synthesized and used for this study. The tubers of *Ipomoea mauritiana* were used as negative control. Briefly, polymerase chain reaction (PCR) was performed by adding 25 ng of plant genomic DNA, 10 mM dNTP mix, 50 pM of SCAR oligonucleotide primer (Pt 1 FP; Pt 1 RP), 2.5 μl of 10× PCR buffer, 2.5 mM of $MgCl_2$, 2.0 U of *Taq* DNA polymerase and making up to 25 μl with distilled water. The tube was placed in a thermocycler (Gradient PCR, Bio-Rad) and PCR was performed as follows: 95°C for 4 min, 35 cycles of 95°C for 45 s, 58°C for 45 s and 72°C for 60 s, and a final extension at 72°C for 10 min. The PCR products were resolved on a 1.2% agarose gel. The DNA was stained with ethidium bromide and photographed under UV trans-illumination. A single, distinct and bright band of 320 bp was obtained with DNA isolated from all the samples of *P. tuberosa* and no non-specific amplification was observed in *Ipomoea mauritiana* (Fig. 1).

Preparation of *Pueraria tuberosa* embedded biscuits

For biscuit preparation, fat and oil, sugar, salt, wheat-flour and PT tuber powder were thoroughly mixed at low speed. The dough was placed on a metal tray and divided into small pieces, slightly flattened by hand, and rolled to a thickness with a rolling pin, using gauge strips. The dough was cut with a biscuit cutter and baked at 205°C for 10 min. Similarly, a standard biscuit without PT tuber powder was also prepared, and used for experimental control animals.

Characterization of PT biscuits

Sensory evaluation of medicinal biscuits was carried out organoleptically for different quality attributes such as colour and appearance, flavour, taste and smell, crispness, and mouth feel by using a 9-point score card (hedonic card). The moisture content, total protein, fat content and ash content were determined by a gravimetric method, micro-Kjeldahl method, Soxhlet extraction method and dry ashing procedure, respectively. The total protein was determined by estimating the total nitrogen. A high-temperature muffle furnace was used in the dry ashing procedure.

Experimental planning and treatment schedule

The protocol of this study was approved by the Institutional Animal Ethical Committee of our Institute of Medical Sciences (BHU, India). The inbred albino mice of Swiss strain (both sexes, 20–30 g) were purchased from the Central Animal Facility of Institute of Medical Sciences, BHU, and acclimatized in our laboratory conditions for 7 days with free access to normal standard chow diet and tap water. Mice were randomly divided into groups I–V, in which group IV and V were again divided into three subgroups, i.e. A, B, C and D, E, F, respectively, with 6 animals in each as given below.

Every day about 18 g of biscuit with 1, 2 or 4 g of PT powder were given to each cage, containing 3 animals. After 7 days of this pretreatment, cisplatin injection (cytoplatin-10) was administered through the intra-peritoneal (i.p.) route to all animals except control (group II) and PT-biscuit experimental control (group IV), but the biscuit administration was continued up 10 days. Blood was collected on the 7th day (before cisplatin injection) and on the 10th day from retro-orbital punctures [20] in a plain tube for serum biochemistry. The animals were killed on the 10th day and kidneys were saved for biochemical studies to assess the antioxidant status of each animal.

Group I (normal control): No treatment.
Group II (Control): Biscuit without PT was given for 10 consecutive days.
Group III (CP experimental control): Biscuit without PT was given for 10 consecutive days but on the 7th day cisplatin (8 mg/kg body weight) was administered i.p.
Group IV (PT-biscuit experimental control):
Group A: Biscuit with 1 g PT powder/18 g biscuit was given for 10 consecutive days
Group B: Biscuit with 2 g PT powder/18 g biscuit was given for 10 consecutive days.
Group C: Biscuit with 4 g PT powder/18 g biscuit was given for 10 consecutive days.
Group V (PT-biscuit-treated animals):
Group D: Biscuit with 1 g PT powder/18 g biscuit was given for 10 consecutive days but on the 7th day cisplatin (8 mg/kg body weight) was administered i.p.
Group E: Biscuit with 2 g PT powder/18 g biscuit was given for 10 consecutive days but on the 7th day cisplatin (8 mg/kg body weight) was administered i.p.
Group F: Biscuit with 4 g PT powder/18 g biscuit as given for 10 consecutive days but on the 7th day cisplatin (8 mg/kg body weight) was administered i.p.

Measurement of biochemical parameters in blood

Hepatic function tests were carried out in terms of serum alanine aminotransferase (ALT), aspartate aminotransferase (AST) [21], total and direct bilirubin (BT and BD) [22], and renal function tests in terms of blood urea nitrogen (BUN) [23] and serum creatinine (CRE) [24] were evaluated by using an auto-analyzer with different methods.

Antioxidant parameters in kidney tissue

Kidney was minced and homogenized in 0.1 M potassium phosphate buffer (pH 7) containing protease inhibitor in a tissue homogenizer to obtain 10% homogenate (w/v), which was centrifuged at 10,000*g* for 20 min at 4°C. The supernatant was utilized for antioxidant status. The level of lipid peroxides was measured as thiobarbituric acid reacting substance (TBARS) and is expressed as equivalent to malondialdehyde (MDA) using 1′,1′,3′,3′-tetramethoxypropane as standard [25, 26]. Superoxide dismutase (SOD) activity was assessed in terms of its ability to inhibit the reduction of nitro blue tetrazolium (NBT) by superoxide, which was instantly generated in the reaction system through a photosensitive reaction in the presence of riboflavin [27, 28]. Reduced glutathione (GSH) level was measured colorimetrically as protein-free sulfhydryl content using 5,5-dithiobis-2-nitrobenzoic acid (DTNB) [29, 30]. Total protein content was determined by the Lowry method using bovine serum albumin as a standard [31].

Statistical analysis

All results were expressed as mean ± standard deviation (SD). The significance of difference among groups was determined using one-way ANOVA followed by Dunnett's post-hoc test. A probability level of less than 5% ($p < 0.05$) was considered significant.

Results

Characterization of PT sample

The PT tuber was first evaluated by texture and microscopic studies. It showed the appearance of dicotyledonous root. The sample was further characterized by RAPD, which showed amplification through PT-specific SCAR primers. It was characterized by TLC fingerprinting in various solvent systems, which produced three spots with R_f value 0.13, 0.38, 0.51 in benzene:ethyl acetate (8:2), four spots with R_f value 0.11, 0.23, 0.51, 0.68 in benzene:ethyl acetate (6:4) and four spots with R_f value 0.26, 0.47, 0.72, 0.87 in benzene:ethyl acetate (2:8).

Characterization of PT biscuit

The PT biscuits showed a gradual decrease in protein and fat content on increasing the concentration of PT powder (Table 1). This change might be due to a comparative decrease in flour content in the biscuit. The moisture and ash content were increased along with increasing PT in the biscuit, which suggested that the PT powder had a water-retaining capacity.

Sensory evaluation indicated gradual enhancement except for sample C (6.44) (Table 1). Sample D had the highest sensory marks, in the range of 8.2, which means that the product was well liked by its users. Samples A and B scored 7.72 and 8.1, respectively. Because sample C (3 g PT biscuit) had poor sensory value it was not included in the biological study.

Table 1 Characterization of PT biscuit (analytical evolution)

Sample no.	Protein (%)	Fat (%)	Moisture (%)	Ash (%)	Sensory marks
A (1 g PT)	7.7	1.5	0.5	0.5	7.7, like moderately
B (2 g PT)	6.8	1.4	0.5	0.5	8.1, like very much
C (3 g PT)	6.6	1.4	0.6	0.6	6.4, like slightly
D (4 g PT)	6.2	1.4	0.6	0.6	8.2, like very much

Effect on body weight

There was a decrease in body weight (recorded on the 7th day), with increase in PT ratio in the PT-biscuit-treated groups, except in 1 g PT biscuits, where no significant decrease was noticed on the 7th day. However, after cisplatin treatment, a decrease in body weight was observed in all the groups (Table 2). A significant decrease in body weight was observed in biscuit experimental control at 2 and 4 g on the 7th and 10th days, which suggests a toxic effect of PT on metabolism.

Blood biochemistry

No change in blood urea nitrogen and creatinine were noted on the 7th day, but they were significantly enhanced on the 10th day in CP experimental control animals. However, PT-treated animals showed significant inhibition in this rise. Those animals who were treated with 4 g PT powder (Table 3) exhibited a 38.2% reduction (from 86.96 ± 11.06 to 53.76 ± 7.55 mg/dl) in urea and a 53.79% reduction (from 1.45 ± 0.06 to 0.67 ± 0.09 mg/dl) in creatinine. However, doses corresponding to PT biscuits of 1 and 2 g did not show any significant protection (Table 3). No change was observed in kidney function tests in animals kept on biscuit experimental control groups on the 10th day.

When their responses were checked on liver function tests, there was no rise in the level of AST and ALT in animals eating 1 g PT biscuits (7th day and 10th day blood samples). However, in similar conditions, higher doses of PT biscuits (2 and 4 g PT-biscuit) only treatment showed raised values of these enzymes, indicating an adverse effect on liver function (Table 4).

Furthermore, when cisplatin was given to these animals, the AST and ALT activities on the 10th day were significantly higher than on the 7th day. Although the 2 g PT biscuit showed a little protection, which was not significant, the 4 g PT biscuit showed enhanced damage to the liver, in addition to cisplatin-induced changes. However, there was no change in BT and BD levels. A significant increase in AST and ALT was observed in biscuit experimental control at 2 and 4 g on the 7th and 10th days, indicating an adverse effect on liver function.

Antioxidant enzymes in kidney homogenate

There was a significant decline in GSH content after cisplatin treatment. On the other hand, SOD activity and TBARS content were significantly higher when compared to normal control rats (Table 5), showing the state of oxidative stress. The 4 g PT biscuit significantly reversed all these changes, although it did not reach normal value on the 10th day of experiment, which was the 3rd day after CP

Table 2 Effect of different dose of biscuits containing *Pueraria tuberosa* powder (1, 2 and 4 g) given for 10 days on the body weight (in g) of mice

Day	Normal control (Group I)	Control (Group II)	CP experimental control (Group III)	PT biscuit experimental control (Group IV)			PT biscuit treated + CP (Group V)		
				1 g PT biscuit experimental control (Group A)	2 g PT biscuit experimental control (Group B)	4 g PT biscuit experimental control (Group C)	1 g PT biscuit + CP (Group D)	2 g PT biscuit + CP (Group E)	4 g PT biscuit + CP (Group F)
0	23.2 ± 0.96	22.2 ± 0.78	25.5 ± 1.9	22.6 ± 2.2	24.5 ± 1.4	22.7 ± 2.1	20.8 ± 1.2	22.5 ± 0.58	22.7 ± 3.0
7	23.0 ± 0.82	24.4 ± 0.62	25.2 ± 2.1	22.5 ± 2.1	21.6 ± 1.3*	19.5 ± 2.5*	20.5 ± 1.9	18.5 ± 0.58*	18.7 ± 1.5*
10	24.3 ± 0.50	23.1 ± 0.77	15.75 ± 2.1**	23.1 ± 2.0	20.7 ± 1.5*	18.5 ± 2.0*	16.0 ± 1.0**	16.7 ± 0.50**	14.5 ± 1.0**

Data are expressed as mean ± SD ($n = 6$ per group); significant differences in groups versus day 0: * $p < 0.05$; ** $p < 0.01$

Normal control: no treatment; control: biscuit only, without PT; CP experimental control: biscuit without PT + cisplatin injection (8 mg/kg body weight). PT biscuit experimental control (1, 2 and 4 g PT biscuit only); PT-biscuit-treated groups: (1, 2 and 4 g PT biscuit) + cisplatin injection (8 mg/kg body weight)

injection. However, the PT biscuits with lower PT content (1 and 2 g) did not show any change by themselves on the 7th day and also failed to show any significant protection against cisplatin-induced changes on the 10th day. The biscuit experimental control groups showed no change in antioxidant status on the 7th and 10th days.

Discussion

The prime objective in preparing the PT biscuit is to develop a herbal food supplement for treatment of adverse effects associated with cisplatin-mediated nephrotoxicity. Cisplatin (CP), a platinum compound, is an potent chemotherapy agent extensively used to treat a variety of malignancies, but its use is compromised because of its associated nephrotoxicity. CP mainly affects the epithelial cells of the proximal tubule, because here its concentration is about five times higher than the serum concentration [32]. We have previously reported the nephroprotective property of a polar fraction of PT tubers (PTY) [8], but hepatotoxicity of higher doses of PT extract has also been shown [17]. Therefore, it becomes important to assess both the properties of PT biscuits. The reduction in body weight in PT-biscuit-treated animals indicates its overall adverse effect, which may be due to hepatic changes in higher doses as described earlier [17].

The results of our study of blood (renal function test) revealed that administration of cisplatin alone produced acute tubular damage, as evidenced by the elevated levels of serum urea and creatinine. Interestingly, treatment with 4 g PT biscuit diminished the elevated serum urea and creatinine, suggesting its reno-protective effect.

Several studies have shown that the cisplatin accumulates in proximal tubules of kidney and then induces oxidative stress, which results in cell damage and compromised renal function [4, 33–36]. The interaction of reactive oxygen species (ROS) with cellular components may damage DNA, proteins and lipids. In support of this theory, several reports are available which show that antioxidants are protective against CP induced nephrotoxicity [4, 37–44]. Therefore, the balance between oxidants and antioxidants is crucial for the maintenance of the biological integrity of the tissues [45]. Thus, the protective effect of PT biscuits may be associated with its antioxidant [15] and anti-inflammatory properties [16], which have been reported earlier.

The reduction of GSH is an early phenomenon occurring in cisplatin-induced lipid peroxidation and subsequent toxicity. Reduced GSH, an endogenous free thiol, functions as a scavenger of ROS, including hydroxyl radicals, singlet oxygen, nitric oxide and peroxynitrile [46]. Here, 4 g PT biscuit treatment significantly prevented the reduction in

Table 3 Effect of different doses of PT biscuits (1, 2 and 4 g) on urea and creatinine

Parameters	Normal control (Group I)		Control (Group II)		CP experimental control (Group III)		PT biscuit experimental control (Group IV)					
							1 g PT biscuit experimental control (Group A)		2 g PT biscuit experimental control (Group B)		4 g PT biscuit experimental control (Group C)	
	D7	D10	D7	D10	D7	D10	D7	D10	D7	D10	D7	D10
Urea (mg/dl)	36.63 ± 2.2	35.65 ± 2.4	34.73 ± 1.2	36.50 ± 2.4	36.30 ± 4.1	86.96 ± 11.1*	36.45 ± 3.1	35.34 ± 2.5	37.22 ± 1.7	35.55 ± 1.4	35.44 ± 2.6	36.27 ± 2.1
Creatinine (mg/dl)	0.48 ± 0.09	0.49 ± 0.06	0.48 ± 0.07	0.47 ± 0.08	0.48 ± 0.09	1.45 ± 0.06*	0.4 ± 0.09	0.48 ± 0.09	0.47 ± 0.09	0.48 ± 0.09	0.48 ± 0.09	0.47 ± 0.09

Parameters	PT biscuit treated + CP (Group V)					
	1 g PT biscuit + CP (Group D)		2 g PT biscuit + CP (Group E)		4 g PT biscuit + CP (Group F)	
	D7	D10	D7	D10	D7	D10
Urea (mg/dl)	36.30 ± 4.1	81.63 ± 6.1#	35.63 ± 5.3	86.70 ± 10.5#	39.23 ± 3.9	53.76 ± 7.6##
Creatinine (mg/dl)	0.45 ± 0.04	1.39 ± 0.07#	0.40 ± 0.08	1.40 ± 0.08#	0.38 ± 0.04	0.67 ± 0.09##

D day

Data are expressed as mean ± SD ($n = 6$ per group); significant differences in control group versus CP experimental group are * $p < 0.05$, ** $p < 0.01$; significant differences in PT-biscuit-treated group versus biscuit experimental controls are # $p < 0.05$, ## $p < 0.01$

Normal control: no treatment; control: biscuit only without PT; CP experimental control: biscuit without PT + cisplatin injection (8 mg/kg body weight). PT biscuit experimental control (1, 2 and 4 g PT biscuit only); PT-biscuit-treated groups: (1, 2 and 4 g PT biscuit) + cisplatin injection (8 mg/kg body weight)

Table 4 Effect of different doses of PT biscuits (1 g, 2 g and 4 g) on liver markers in blood samples of mice

Parameters	Normal control (Group I)		Control (Group II)		CP experimental control (Group III)		PT biscuit experimental control (Group IV)					
							1 g PT biscuit experimental control (Group A)		2 g PT biscuit experimental control (Group B)		4 g PT biscuit experimental control (Group C)	
	D7	D10	D7	D10	D7	D10	D7	D10	D7	D10	D7	D10
AST (IU/L)	67.1 ± 10.0	65.0 ± 12.0	65.2 ± 11.0	69.1 ± 11.1	56.3 ± 12.2	104.0 ± 10.0*	58.1 ± 3.4	56.2 ± 2.8	105.7 ± 4.1*	128.8 ± 8.2*	106.3 ± 5.8*	116.7 ± 7.1*
ALT (IU/L)	181.2 ± 14.6	176.3 ± 17.7	180.3 ± 13.4	179.4 ± 19.9	174.6 ± 22.4	230.8 ± 15.6*	171.1 ± 10.0	168.2 ± 8.8	213.0 ± 8.7*	245.2 ± 6.1	196.2 ± 7.5**	208.6 ± 13.8*
BT (mg/dl)	0.40 ± 0.10	0.42 ± 0.07	0.40 ± 0.10	0.42 ± 0.07	0.40 ± 0.10	0.40 ± 0.12	0.40 ± 0.10	0.40 ± 0.10	0.40 ± 0.10	0.40 ± 0.13	0.40 ± 0.13	0.40 ± 0.10
BD (mg/dl)	0.17 ± 0.05	0.16 ± 0.05	0.17 ± 0.05	0.16 ± 0.05	0.15 ± 0.05	0.16 ± 0.04	0.15 ± 0.04	0.16 ± 0.04	0.15 ± 0.04	0.16 ± 0.04	0.15 ± 0.04	0.16 ± 0.04

Parameters	PT biscuit treated + CP (Group V)					
	1 g PT biscuit + CP (Group D)		2 g PT biscuit + CP (Group E)		4 g PT biscuit + CP (Group F)	
	D7	D10	D7	D10	D7	D10
AST (IU/L)	55.1 ± 5.6	116.5 ± 11.9*	113.4 ± 6.1*	99.1 ± 7.1*	112.6 ± 6.8*	131.7 ± 8.1*
ALT (IU/L)	167.8 ± 8.11	248.7 ± 22.6*	237.0 ± 7.7*	195.2 ± 5.1	216.2 ± 9.4*	238.6 ± 17.5*
BT (mg/dl)	0.40 ± 0.10	0.40 ± 0.05	0.40 ± 0.08	0.40 ± 0.13	0.40 ± 0.10	0.40 ± 0.09
BD (mg/dl)	0.15 ± 0.04	0.16 ± 0.04	0.15 ± 0.04	0.16 ± 0.04	0.15 ± 0.04	0.16 ± 0.05

Data are expressed as mean ± SD ($n = 6$ per group); significant differences in groups versus control group were as follows * $p < 0.05$; ** $p < 0.01$

Normal control: no treatment; control: biscuit only without PT; CP experimental control: biscuit without PT + cisplatin injection (8 mg/kg body weight). PT biscuit experimental control (1, 2 and 4 g PT biscuit only); PT-biscuit-treated groups: 1, 2 and 4 g PT biscuit + cisplatin injection (8 mg/kg body weight)

AST aspartate aminotransferase, *ALT* alanine aminotransferase (ALT), *BT* total bilirubin, *BD* direct bilirubin, *D* day

Table 5 Protective effect of PT biscuit on levels of antioxidant parameters in kidney tissue homogenate isolated from rats treated with cisplatin on 10th day

Group	GSH (μg/mg protein)	SOD (U/mg protein)	MDA (nmol/mg protein)
Normal control (Group I)	8.04 ± 0.45	15.16 ± 0.42	0.135 ± 0.03
Control (Group II)	7.64 ± 0.55	14.58 ± 0.48	0.128 ± 0.03
CP experimental control (Group III)	3.98 ± 0.15***	8.32 ± 0.29***	0.355 ± 0.03**
PT biscuit experimental control (Group IV)			
1 g PT biscuit experimental control (Group A)	8.54 ± 0.48	15.31 ± 0.52	0.137 ± 0.03
2 g PT biscuit experimental control (Group B)	7.55 ± 0.39	15.68 ± 0.46	0.131 ± 0.02
4 g PT biscuit experimental control (Group C)	8.37 ± 0.55	14.81 ± 0.54	0.125 ± 0.03
PT biscuit treated (Group V)			
1 g PT biscuit + CP (Group D)	$3.92 \pm 0.11^{\#\#\#}$	$7.75 \pm 0.22^{\#\#\#}$	$0.368 \pm 0.02^{\#\#}$
2 g PT biscuit + CP (Group E)	$3.49 \pm 0.13^{\#\#\#}$	$7.99 \pm 0.24^{\#\#\#}$	$0.328 \pm 0.03^{\#\#}$
4 g PT biscuit + CP (Group F)	$5.17 \pm 0.15^{\#}$	$11.42 \pm 0.56^{\#}$	$0.218 \pm 0.03^{\#}$

Data are expressed as mean ± SD ($n = 6$ per group); significant differences in CP experimental group versus control group are * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; significant differences in PT-biscuit-treated group versus biscuit experimental controls are $^{\#}$ $p < 0.05$, $^{\#\#}$ $p < 0.01$, $^{\#\#\#}$ $p < 0.001$

Normal control: no treatment; control: biscuit only without PT; CP experimental control: biscuit without PT + cisplatin injection (8 mg/kg body weight). PT biscuit experimental control: 1, 2 and 4 g PT biscuit only; PT-biscuit-treated groups: 1, 2 and 4 g PT biscuit + cisplatin injection (8 mg/kg body weight)

reduced GSH content in kidney tissue homogenate. Excess free radicals induce lipid peroxidation, which has been monitored here as TBARS. Interestingly, our data showed a significant increase in TBARS in the CP-treated group, which was significantly reversed in PT-biscuit treated animals. Similar, PT-mediated prevention has also been observed in the case of CP-induced changes in SOD activity, which is an endogenous enzymatic defence system to protect against free radical stress. There was decrease in SOD activity in CP-treated animals, which was prevented in 4 g PT-biscuit-treated animals.

Our earlier report indicated rises in serum ALT and AST, without significant changes in serum BT and BD in PTME (methanolic extract of PT)-treated animals [17], which indicates no effect on the hepatobiliary system [47]. Interesting, the PT powder, which is rich in fibers and glycosides [10], also showed similar hepatotoxicity when given in equivalent doses (calculated on the basis of % yield; data not shown). In a parallel experiment, CP treatment alone also showed similar hepatotoxicity [48], but when PT and CP were given simultaneously, there was an additive response on liver function enzymes but an opposite effect on kidney function enzymes.

In our data, PT biscuits with 2 and 4 g showed 16 and 40% enhancement in AST and ALT, respectively, in mice on the 7th day, but with cisplatin 4 g PT biscuit showed a kind of additive response for AST and ALT enhancement. This hepatotoxicity might be due to one of the secondary metabolites in PT, which could be present in very low concentration but show a significant effect when taken in higher doses. The traditional ayuvedic text also advocates its use in lower doses only, because its higher doses have been associated with inducing a tendency to vomit [10].

Conclusion

Based on the above results, it could be concluded that PT biscuits are a good food supplement to prevent cisplatin-mediated oxidative stress and kidney damage, but should be recommended either as intermittent therapy or in lower doses. It is therefore advised to use PT biscuits with caution, with continuous monitoring of liver function, and they should be discontinued after a certain period of administration.

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